

PBA: 13. Hypothesis Testing

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Roadmap

1. The Lady Tasting Tea
2. Hypothesis Tests Using `infer`
3. Two-Sample Permutation Tests with `infer`
4. Power Analysis
5. Issues with Hypothesis Testing

1/ The Lady Tasting Tea

The Lady Tasting Tea



Image created using DALL·E 3 depicting R.Fisher's experiment.

“Your friend asks you to grab a tea with milk for her before meeting up and she says that she prefers tea poured before the milk. You stop by a local tea shop and ask for a tea with milk. When you bring it to her, she complains that it was prepared milk-first.”

- You're skeptical that she can tell the difference, so you devise a test:
 - Prepare 8 cups of tea, 4 milk-first, 4 tea-first.
 - Present cups to friend in a **random** order.
 - Ask friend to pick which 4 of the 8 were milk-first.

Lady Tasting Tea Data

Results from the test / experiment:

```
1 # don't forget to load tidyverse and dplyr as usual!
2 > tea <- read_csv("https://bit.ly/3RfqcGz"); tea
3
4 # A tibble: 8 × 2
5   truth      guess
6   <chr>      <chr>
7 1 tea-first tea-first
8 2 milk-first milk-first
9 3 milk-first milk-first
10 4 tea-first tea-first
11 5 tea-first tea-first
12 6 milk-first milk-first
13 7 tea-first tea-first
14 8 milk-first milk-first
```

Friend picks out all 4 milk-first cups correctly!

```
1 > tea |>
2   summarize(sum(truth == guess))
3
4 # A tibble: 1 × 1
5   `sum(truth == guess)`
6   <int>
7 1             8
```

Thought Experiment

Could she have been guessing at random? What would guessing look like?

```
1 > set.seed(9999)
2 > one_guess <- tea |>
3   mutate(random_guess = sample(guess)); one_guess
4
5 # A tibble: 8 × 3
6   truth    guess    random_guess
7   <chr>    <chr>    <chr>
8 1 tea-first tea-first tea-first
9 2 milk-first milk-first milk-first
10 3 milk-first milk-first tea-first
11 4 tea-first tea-first milk-first
12 5 tea-first tea-first milk-first
13 6 milk-first milk-first milk-first
14 7 tea-first tea-first tea-first
15 8 milk-first milk-first tea-first
```

4 correct in this random guess!

```
1 > one_guess |>
2   summarize(sum(truth == random_guess))
3
4 # A tibble: 1 × 1
5   `sum(truth == random_guess)`
6   <int>
7 1 4
```

Another Guess

6 correct in this random guess!

```
1 > another_guess <- tea |>
2   mutate(random_guess = sample(guess)); another_guess
3
4 # A tibble: 8 × 3
5   truth      guess      random_guess
6   <chr>    <chr>    <chr>
7 1 tea-first tea-first milk-first
8 2 milk-first milk-first milk-first
9 3 milk-first milk-first tea-first
10 4 tea-first tea-first tea-first
11 5 tea-first tea-first tea-first
12 6 milk-first milk-first milk-first
13 7 tea-first tea-first tea-first
14 8 milk-first milk-first milk-first
15
16 > another_guess |>
17   summarize(sum(truth == random_guess))
18
19 # A tibble: 1 × 1
20   `sum(truth == random_guess)`
21                               <int>
22 1                               6
```

All Possible Guesses

We could enumerate all possible guesses:

##	Cup 1	Cup 2	Cup 3	Cup 4	Cup 5	Cup 6	Cup 7	Cup 8
## 1	milk	milk	milk	milk	tea	tea	tea	tea
## 2	milk	milk	milk	tea	milk	tea	tea	tea
## 3	milk	milk	tea	milk	milk	tea	tea	tea
## 4	milk	tea	milk	milk	milk	tea	tea	tea
## 5	tea	milk	milk	milk	milk	tea	tea	tea
## 6	milk	milk	milk	tea	tea	milk	tea	tea

[snip]

##	Cup 1	Cup 2	Cup 3	Cup 4	Cup 5	Cup 6	Cup 7	Cup 8
## 65	tea	tea	tea	milk	milk	tea	milk	milk
## 66	milk	tea	tea	tea	milk	milk	milk	milk
## 67	tea	milk	tea	tea	milk	milk	milk	milk
## 68	tea	tea	milk	tea	milk	milk	milk	milk
## 69	tea	tea	tea	milk	tea	milk	milk	milk
## 70	tea	tea	tea	tea	milk	milk	milk	milk

↪ “Guessing” would mean choosing one of these at random!

Statistical Thought Experiment

- Statistical thought experiment: how often would she get all 4 correct **if she were guessing randomly?**
 - Only one way to choose all 4 correct cups.
 - But 70 ways of choosing 4 cups among 8.
 - Choosing at random: picking each of these 70 with equal probability.
- Chances of guessing all 4 correct is $\frac{1}{70} \approx 0.014$ or 1.4%.
 - $\rightsquigarrow$ the guessing hypothesis might be implausible.
 - Impossible? No, because of random chance!

2/ Hypothesis Tests Using **infer**

Statistical Hypothesis Testing

- Statistical hypothesis testing is a **thought experiment**.
 - Could our results just be due to random chance?
- What would the world look like **if we knew the truth**?
- Example 1:
 - An industry analyst claims 39% of market demand for Tesla's Cybertruck.
 - The GLG market research poll reports that 43% of their sample is interested in purchasing (Week 12 on Polls).
 - Q: Should you conclude that the analyst is wrong?
- Example 2:
 - Trump won 47.5% of the votes in the 2020 election.
 - Last [YouGov](#) poll of 1,363 likely voters said 44% planned to vote for Trump.
 - Q: Could the difference between the poll and the outcome just be due to random chance?

Null and Alternative Hypothesis

- **Null hypothesis:** Some statement about the population parameters.
 - “Devil’s advocate” position $\rightsquigarrow$ assumes what you seek to prove wrong.
 - Claims that an observed difference is due to random chance.
 - Ex: YouGov poll was drawn from the same population as all voters.
 - Denoted H_0
- **Alternative hypothesis:** The statement we hope (or suspect) is true instead of H_0 .
 - The opposite of the null hypothesis.
 - An observed difference is real, not just due to random chance.
 - Ex: YouGov polling for Trump was systematically wrong.
 - Denoted H_1 or H_a
- **Probabilistic** proof by contradiction: try to “disprove” the null.

Hypothesis Testing Example

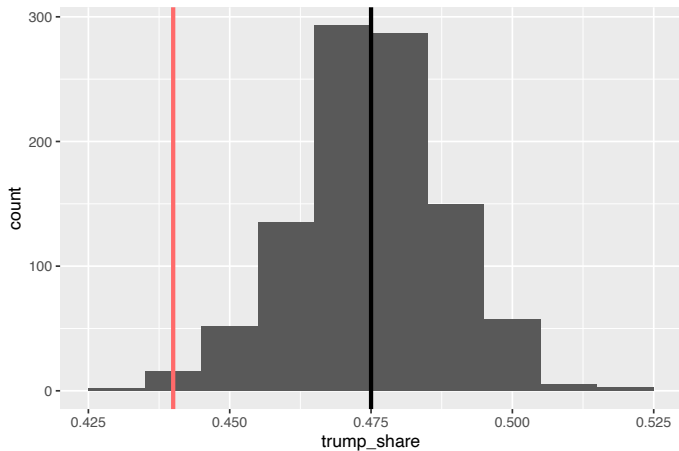
- Returning to the YouGov Poll example:
 - Are we polling the same population as the actual voters? (H_0)
 - If so, how likely are we to see polling error this big by random chance?
↪ For this, imagine having access to multiple polls (repeated samples)!
- What is the parameter we want to learn about?
 - True population mean of the polls/surveys, p .
 - Null hypothesis: $H_0 : p = 0.475$ (i.e., polls drawing from the same population)
 - Alternative hypothesis: $H_1 : p \neq 0.475$
- Data: YouGov poll has $\bar{X} = 0.44$ with $n = 1,363$.

Statistical Thought Experiment

- If the null were true, what should the distribution of the data be?
 - X_i is 1 if respondent i will vote for Trump.
 - Under null, X_i is a coin flip with probability $p = 0.475$ of “landing on Trump”.
 - $X_1 + X_2 + \dots + X_n$ is the number in sample that will vote for Trump.
- We can simulate sums of coin flips using a function called `rbinom()`.
- Compare the distribution of proportions under the null to the observed proportion.
 - ↪ “reference distribution” as we will be comparing it to the observed quantity/data!

```
1 > null_dist <- tibble(  
2   trump_share = rbinom(n = 1000, size = 1363, prob = 0.475) / 1363  
3 )  
4  
5 > ggplot(null_dist, aes(x = trump_share)) +  
6   geom_histogram(binwidth = 0.01) +  
7   geom_vline(xintercept = 0.44, color = "indianred1", size = 1.25) +  
8   geom_vline(xintercept = 0.475, size = 1.25)
```

Simulation of the Reference Distribution



P-Value

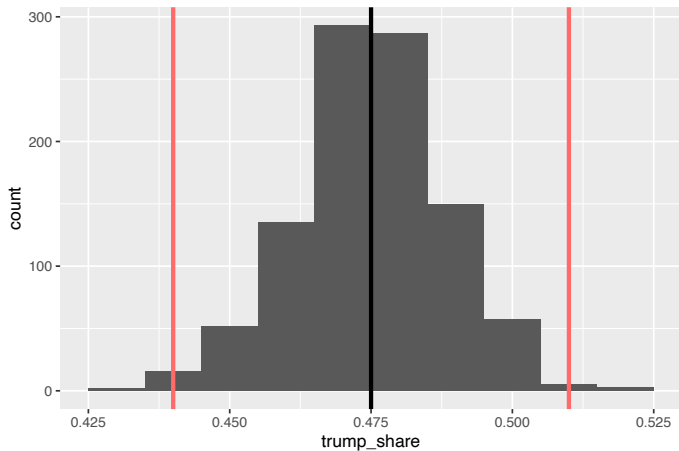
p-value

The p-value is the probability of observing data as or more extreme as our data if the null hypothesis is true.

- If the null is true, how often would we expect polling errors this large?
 - Smaller p-value $\Rightarrow$ stronger evidence against the null
 - Caveat: **NOT** the probability that the null is true!
- p-values are usually **two-sided**:
 - Observed error of $0.44 - 0.475 = -0.035$ under the null.
 - the (two-sided) p-value =
 - Probability of sample proportions being less than 0.44 **plus**
 - Probability of sample proportions being greater than 0.51 ($= 0.475 + 0.035$).

```
1 > mean(null_dist$trump_share < 0.44) + mean(null_dist$trump_share > 0.51)
2 [1] 0.008
```

Simulation of the Reference Distribution



One-Sided Tests

- Sometimes our hypothesis is directional.
 - We only consider evidence against the null from one direction.
- Null: our polls are from the same population as actual voters.
 - $H_0 : p = 0.475$
- **One-sided alternative:** polls underestimate Trump support.
 - $H_1 : p < 0.475$
- Makes the p-value one-sided:
 - What's the probability of a random sample underestimating Trump support by as much as we see in the sample (e.g., 3.5%)?
 - Always smaller than a two-sided p-value! Q: Why?

```
1 > mean(null_dist$trump_share < 0.44)
2 [1] 0.005
```

Rejecting the Null

- Tests usually end with a decision to reject the null or not.
- We will need to choose a **threshold** below which you'll reject the null.
 - Test level α : the threshold for a test.
 - Decision rule: "reject the null if the p-value is below α "
 - Otherwise "fail to reject" or "retain", **not "accept the null"**
- Common (arbitrary) thresholds:
 - $p \geq 0.1$ "not statistically significant"
 - $p < 0.05$ "statistically significant"
 - $p < 0.01$ "highly significant"

Testing Errors

- A p-value of 0.05 says that data this extreme would only happen in 5% of repeated samples if the null were true.
 - $\rightsquigarrow$ 5% of the time we'll reject the null when it is actually true.
- Test errors:

	H_0 True	H_0 False
Retain H_0	Awesome!	Type II error
Reject H_0	Type I error	Good stuff!

- Q: Which test errors are more serious?
 - Type I error!
 - “Convicting” an innocent null hypothesis.
 - e.g., drug not effective but we claim it is.
 - Type II error less serious:
 - Missed out on an awesome finding.

The infer Workflow for Hypothesis Testing

GSS Data from infer

```
1 > library(infer) # if you haven't done so already!
2 > gss # Subset of data from the General Social Survey (GSS).
3
4 # A tibble: 500 × 11
5   year age sex college partyid hompop hours income class finrela weight
6   <dbl> <dbl> <fct> <fct> <fct> <dbl> <dbl> <ord> <fct> <fct> <dbl>
7 1 2014 36 male degree ind 3 50 $25000 or more middle c below 0.896
8 2 1994 34 female no degree rep 4 31 $20000 - 24999 working below 1.08
9 3 1998 24 male degree ind 1 40 $25000 or more working below 0.550
10 4 1996 42 male no degree ind 4 40 $25000 or more working above 1.09
11 5 1994 31 male degree rep 2 40 $25000 or more middle c above 1.08
12 6 1996 32 female no degree rep 4 53 $25000 or more middle c average 1.09
13 7 1990 48 female no degree dem 2 32 $25000 or more working below 1.06
14 8 2016 36 female degree ind 1 20 $25000 or more middle c above 0.478
15 9 2000 30 female degree rep 5 40 $25000 or more middle c average 1.10
16 10 1998 33 female no degree dem 2 40 $15000 - 19999 working far be 0.550
17 # 490 more rows
18 # Use `print(n = ...)` to see more rows
```

The General Social Survey (GSS): a long-standing biannual sociological survey that collects data on American society's changing attitudes and behaviors (e.g., working hours, longevity, happiness).

What is the Average Hours Worked?

The dplyr way:

```
1 > gss |>
2   summarize(mean(hours))
3
4 # A tibble: 1 × 1
5   `mean(hours)`
6   <dbl>
7 1      41.4
```

The infer way:

```
1 > observed_mean <- gss |>
2   specify(response = hours) |>
3   calculate(stat = "mean"); observed_mean
4
5 Response: hours (numeric)
6 # A tibble: 1 × 1
7   stat
8   <dbl>
9 1  41.4
```

Devise a Hypothesis Test

Could we get a mean this different from 40 hours if that was the true population average of hours worked?

Null and alternative:

$$H_0 : \mu_{\text{hours}} = 40$$

$$H_1 : \mu_{\text{hours}} \neq 40$$

How do we perform this test using infer? The **bootstrap**!

The infer Workflow

- Statistical hypothesis testing is a **thought experiment**.
- What would the world look like **if we knew the truth**?
- Conducted with several steps:
 1. Specify your **null** and **alternative hypotheses**.
 2. Choose an appropriate **test statistic** (e.g., sample mean) and level of test α .
 3. Derive the **reference/null distribution** of the test statistic under the null.
 4. Use this distribution to calculate the **p-value**.
 5. Use p-value to **decide** whether to reject the null hypothesis or not.

Specify the Hypotheses

```
1 > gss |>
2   specify(response = hours) |>
3   hypothesize(null = "point", mu = 40)
4
5 Response: hours (numeric)
6 Null Hypothesis: point
7 # A tibble: 500 × 1
8   hours
9   <dbl>
10  1    50
11  2    31
12  3    40
13  4    40
14  5    40
15  6    53
16  7    32
17  8    20
18  9    40
19 10    40
20 # 490 more rows
21 # Use `print(n = ...)` to see more rows
```

- Point (vs. independence) hypotheses:
 - ↪ Set hypothesis = "point" to indicate we are testing a specific value (point estimate) of a population parameter, such as the mean working hours being equal to 40.

Generate the Null/Reference Distribution

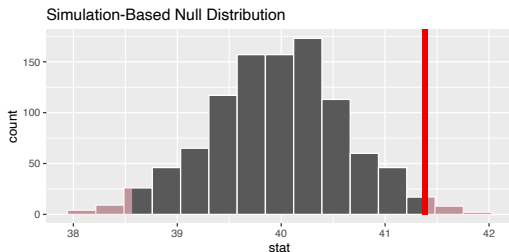
We can use the bootstrap to determine how much variation there will be around 40 in the null distribution.

```
1 > null_dist <- gss |>
2   specify(response = hours) |>
3   hypothesize(null = "point", mu = 40) |>
4   generate(reps = 1000, type = "bootstrap") |>
5   calculate(stat = "mean"); null_dist
6
7 Response: hours (numeric)
8 Null Hypothesis: point
9 # A tibble: 1,000 × 2
10   replicate stat
11   <int> <dbl>
12     1      1 40.8
13     2      2 40.1
14     3      3 39.1
15     4      4 38.8
16     5      5 39.5
17     6      6 39.6
18     7      7 40.2
19     8      8 40.5
20     9      9 39.2
21    10     10 39.8
22 # 990 more rows
23 # Use `print(n = ...)` to see more rows
```

Visualize the P-Value

We can visualize our bootstrapped null distribution and the p-value as a shaded region:

```
1 > null_dist |> visualize() +  
2   shade_p_value(observed_mean, direction = "two-sided")  
3  
4 > null_dist |> get_p_value(obs_stat = observed_mean, direction = "two-sided") |> pull()  
5 [1] 0.032
```



↪ Q: The two-sided p-value is 0.032. If we consider the test level α of .05, do we reject the null?

3/ Two-Sample Permutation Tests with **infer**

Social Pressure Experiment

- Experimental study where each household for 2006 Michigan primary election was randomly assigned to one of 5 conditions:
 - Control: no mail.
 - Civic Duty: a mail saying voting is your civic duty.
 - Hawthorne: a “we’re watching you” message.
 - Self: a mail showing your own voting record as a reminder of your past turnout.
 - Neighbors: a mail naming-and-shaming the recipient for not voting (social pressure).
- Outcome: whether household members voted or not.
- We’ll focus on **Neighbors** vs **Control** comparison.

Treatment Condition: Neighbors

Dear Registered Voter:

WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse. This year, we're taking a new approach. We're sending this mailing to you and your neighbors to publicize who does and does not vote.

The chart shows the names of some of your neighbors, showing which have voted in the past. After the August 8 election, we intend to mail an updated chart. You and your neighbors will all know who voted and who did not.

DO YOUR CIVIC DUTY — VOTE!

MAPLE DR	Aug 04	Nov 04	Aug 06
9995 JOSEPH JAMES SMITH	Voted	Voted	_____
9995 JENNIFER KAY SMITH		Voted	_____
9997 RICHARD B JACKSON		Voted	_____
9999 KATHY MARIE JACKSON		Voted	_____
9999 BRIAN JOSEPH JACKSON		Voted	_____

Source: Gerber, Greene, Larimer (2008)

[<https://doi.org/10.1017/S000305540808009X>]

Social Pressure Data

```
1 > library(infer)
2 > data(social, package = "qss") # also available at "https://bit.ly/4ampWhU"
3 > social <- as_tibble(social); social
4
5 # A tibble: 305,866 × 6
6   sex      yearofbirth primary2004 messages primary2006 hhsz
7   <chr>      <int>      <int> <chr>      <int> <int>
8 1 male      1941          0 Civic Duty      0      2
9 2 female    1947          0 Civic Duty      0      2
10 3 male      1951          0 Hawthorne      1      3
11 4 female    1950          0 Hawthorne      1      3
12 5 female    1982          0 Hawthorne      1      3
13 6 male      1981          0 Control        0      3
14 7 female    1959          0 Control        1      3
15 8 male      1956          0 Control        1      3
16 9 female    1968          0 Control        0      2
17 10 male     1967          0 Control        0      2
18 # 305,856 more rows
19 # Use `print(n = ...)` to see more rows
```

- Data source: Kosuke Imai. 2017. Quantitative Social Science: An Introduction. Princeton University Press. ↪ one of our recommended textbooks!

Two-Sample Hypotheses

- Parameter: **population ATE** $\mu_T - \mu_C$
 - μ_T : Turnout rate in the population if everyone received treatment.
 - μ_C : Turnout rate in the population if everyone received control.
- Goal: learn about the population difference in means.
- Usual null hypothesis: no difference in population means (ATE = 0)
 - Null: $H_0 : \mu_T - \mu_C = 0$
 - Two-sided alternative: $H_0 : \mu_T - \mu_C \neq 0$
- In words: are the differences in sample means just due to random chance?

Permutation Test

- Q: How do we generate draws of the difference in means under the null ($H_0 : \mu_T - \mu_C = 0$)?
- **Key idea:** If the voting distribution is the same in the treatment and control groups, we could randomly swap who is labelled as treated and who is labelled as control and it shouldn't matter!
 - ↪ True if the outcome distributions are **independent** of treatment assignment (i.e., $\mathbb{P}(Y|D = 1) = \mathbb{P}(Y|D = 0)$, thus ATE = 0).
- **Permutation test:** generate the null distribution by permuting the group labels and see the resulting distribution of differences in means.
 - Also referred to as randomization test.

Permuting the Labels

- Use `sample()` to permute the labels with `mutate()`:

```
1 > social <- social |>
2   filter(messages %in% c("Neighbors", "Control"))
3
4 > social |>
5   mutate(messages_permute = sample(messages)) |>
6   select(primary2006, messages, messages_permute)
7
8 # A tibble: 229,444 × 3
9   primary2006 messages messages_permute
10    <int> <chr> <chr>
11     1 0 Control Neighbors
12     2 1 Control Control
13     3 1 Control Control
14     4 0 Control Control
15     5 0 Control Neighbors
16     6 1 Control Control
17     7 0 Control Neighbors
18     8 1 Control Control
19     9 1 Control Control
20    10 1 Control Control
21 # 229,434 more rows
22 # Use `print(n = ...)` to see more rows
```

Calculating the Difference in Proportion

infer functions with binary outcome (i.e., voted or not) work best with factor variables.

- Use the argument `success = level` in `specify()` to indicate the level of response/outcome considered as success (voted = 1).

```
1 > social <- social |>
2   mutate(turnout = if_else(primary2006 == 1, "Voted", "Didn't Vote"))
3
4 > est_ate <- social |>
5   specify(turnout ~ messages, success = "Voted") |>
6   calculate(stat = "diff in props", order = c("Neighbors", "Control")); est_ate
7
8 Response: turnout (factor)
9 Explanatory: messages (factor)
10 # A tibble: 1 × 1
11   stat
12   <dbl>
13 1 0.0813
```

- Our point estimate for the ATE = 0.0813; i.e., diff in proportions between T and C.
- $\rightsquigarrow$ Let's now take a look at how to run permutation test step by step!

Specify the Relationship of Interest

Again, infer functions with binary outcome (i.e., voted or not) work best with factor variables $\rightsquigarrow$ set success = "Voted" in `specify()`:

```
1 > social |>
2   specify(turnout ~ messages, success = "Voted")
3
4 Response: turnout (factor)
5 Explanatory: messages (factor)
6 # A tibble: 229,444 × 2
7   turnout      messages
8   <fct>      <fct>
9 1 Didn't Vote Control
10 2 Voted      Control
11 3 Voted      Control
12 4 Didn't Vote Control
13 5 Didn't Vote Control
14 6 Voted      Control
15 7 Didn't Vote Control
16 8 Voted      Control
17 9 Voted      Control
18 10 Voted     Control
19 # 229,434 more rows
20 # Use `print(n = ...)` to see more rows
```

Set Up the Hypotheses

The null for these two-sample tests is called “independence” for the `infer` package because the assumption is that the two variables are statistically independent (i.e., $Y^{\text{obs}} \perp\!\!\!\perp D$).

```
1 > social |>
2   specify(turnout ~ messages, success = "Voted") |>
3   hypothesize(null = "independence")
4
5 Response: turnout (factor)
6 Explanatory: messages (factor)
7 Null Hypothesis: independence
8 # A tibble: 229,444 × 2
9   turnout      messages
10  <fct>      <fct>
11 1 Didn't Vote Control
12 2 Voted      Control
13 3 Voted      Control
14 4 Didn't Vote Control
15 5 Didn't Vote Control
16 6 Voted      Control
17 7 Didn't Vote Control
18 8 Voted      Control
19 9 Voted      Control
20 10 Voted     Control
21 # 229,434 more rows
22 # Use `print(n = ...)` to see more rows
```

Generate the Permutations

We can tell `infer` to do our permutation test by using the argument `type = "permute"` w/in `generate()`:

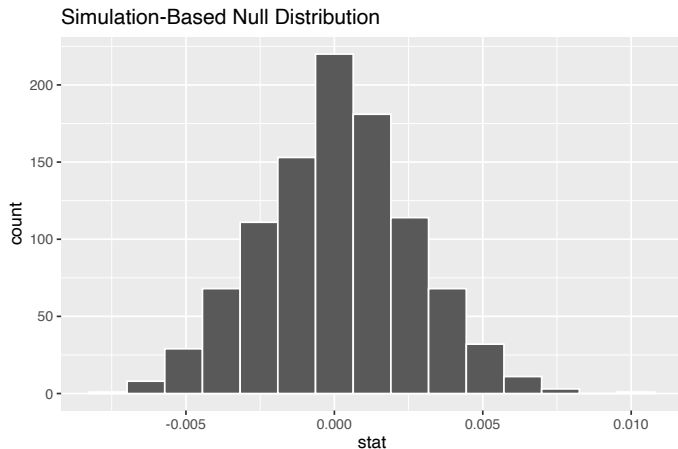
```
1 > social |>
2   specify(turnout ~ messages, success = "Voted") |>
3   hypothesize(null = "independence") |>
4   generate(reps = 1000, type = "permute")
5
6 Response: turnout (factor)
7 Explanatory: messages (factor)
8 Null Hypothesis: independence
9 # A tibble: 229,444,000 × 3
10 # Groups:   replicate [1,000]
11   turnout      messages replicate
12   <fct>      <fct>      <int>
13 1 Didn't Vote Control      1
14 2 Didn't Vote Control      1
15 3 Didn't Vote Control      1
16 4 Voted      Control      1
17 5 Didn't Vote Control      1
18 6 Didn't Vote Control      1
19 7 Didn't Vote Control      1
20 8 Didn't Vote Control      1
21 9 Didn't Vote Control      1
22 10 Voted      Control      1
23 # 229,443,990 more rows
24 # Use `print(n = ...)` to see more rows
```

Calculating the Diff in Props in Each Sample

```
1 > null_dist <- social |>
2   specify(turnout ~ messages, success = "Voted") |>
3   hypothesize(null = "independence") |>
4   generate(reps = 1000, type = "permute") |>
5   calculate(stat = "diff in props", order = c("Neighbors", "Control")); null_dist
6
7 Response: turnout (factor)
8 Explanatory: messages (factor)
9 Null Hypothesis: independence
10 # A tibble: 1,000 × 2
11   replicate    stat
12   <int>      <dbl>
13 1         1  0.00364
14 2         2  0.00470
15 3         3 -0.00252
16 4         4  0.00462
17 5         5  0.00176
18 6         6  0.000721
19 7         7  0.00298
20 8         8 -0.00412
21 9         9  0.00276
22 10        10  0.00336
23 # 990 more rows
24 # Use `print(n = ...)` to see more rows
```

Visualize the Null/Reference Distribution

```
1 > null_dist |>  
2   visualize()
```



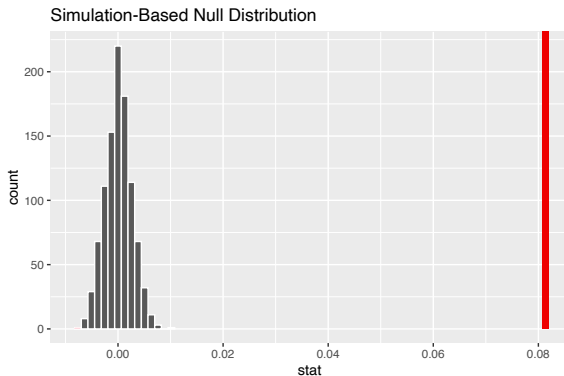
Calculating P-Values

```
1 > ate_pval <- null_dist |>
2   get_p_value(obs_stat = est_ate, direction = "both"); ate_pval
3
4 Warning message:
5 Please be cautious in reporting a p-value of 0. This result is an approximation based on
6 the number of `reps` chosen in the `generate()` step. See `?get_p_value()` for more information.
7
8 # A tibble: 1 × 1
9   p_value
10   <dbl>
11 1      0
```

- Theoretically, a p-value is strictly positive (defined on the interval $(0,1]$).
 - However, `get_p_value()` may return 0 in some cases due to the simulation-based nature of the `infer` package.

Visualize P-Values

```
1 > null_dist |>  
2   visualize() +  
3   shade_p_value(obs_stat = est_ate, direction = "both")
```



- Red vertical line depicts the observed diff. in proportion (i.e., 0.0813; p.33)

Hypothesis Tests and Confidence Intervals

- There is a deep connection between confidence intervals and tests.

```
1 > social |>
2   specify(turnout ~ messages, success = "Voted") |>
3   generate(reps = 1000, type = "bootstrap") |>
4   calculate(stat = "diff in props",
5             order = c("Neighbors", "Control")) |>
6   get_ci(level = 0.95)
7
8 # A tibble: 1 × 2
9   lower_ci upper_ci
10    <dbl>    <dbl>
11 1  0.0766  0.0865
```

- Any value outside of a $100 \times (1 - \alpha)\%$ confidence interval would have a p-value less than α (e.g., 0.05) if we tested it as the null hypothesis.
 - 95% CI for the social pressure experiment: [0.0766, 0.0865]
 - $\rightsquigarrow$ p-value for $H_0 : \mu_T - \mu_C = 0$ less than 0.05.

4/ Power Analysis

Effect Sizes

TABLE 2. Effects of Four Mail Treatments on Voter Turnout in the August 2006 Primary Election

	Experimental Group				
	Control	Civic Duty	Hawthorne	Self	Neighbors
Percentage Voting	29.7%	31.5%	32.2%	34.5%	37.8%
N of Individuals	191,243	38,218	38,204	38,218	38,201

Source: Gerber, Greene, Larimer (2008) [<https://doi.org/10.1017/S000305540808009X>]

- Why did Gerber, Green, and Larimer use sample sizes of 38,000 for each treatment condition?
- Choose the sample size to ensure that you can detect what you think might be the true treatment effect:
 - Small effect sizes (half percentage point) will require huge n
 - Large effect sizes (10 percentage points) will require smaller n
- **Detect** here means “reject the null of no effect”

Power of a Test

- **Definition:** The power of a test is the probability that a test rejects the null.
 - Probability that we reject given some specific value of the parameter.
 - $\text{Power} = 1 - \mathbb{P}(\text{Type II error})$
 - Better tests have higher power.
- If we fail to reject a null hypothesis, two possible states of the world:
 - Null is true (no treatment effect)
 - Null is false (there is a treatment effect), but the test had low power.

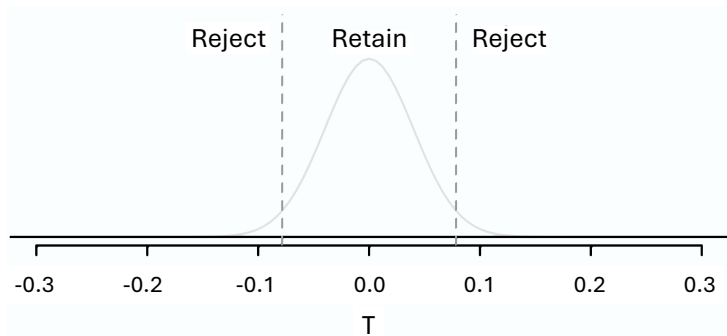
Why Care about Power?

- Imagine you are a company being sued for racial discrimination in hiring.
- Judge forces you to conduct a hypothesis test:
 - Null hypothesis is that hiring rates for white and black applicants are equal, $H_0 : \mu_W - \mu_B = 0$
 - You sample 10 hiring records from each group, run the test, and fail to reject the null.
- Say to judge, “look, we don’t have any racial discrimination!” $\rightsquigarrow$ What’s the problem?

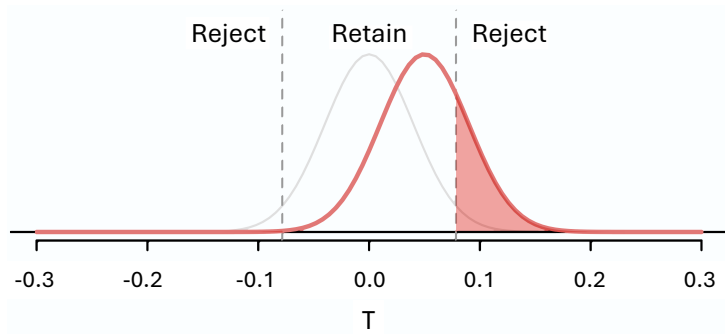
Power Analysis Procedure

- Power can help guide the choice of sample size through a **power analysis**.
 - Calculate how likely we are to reject different possible treatment effects at different sample sizes.
 - **Can be done before the experiment:** which effects will I be able to detect with high probability at my n ?
- Steps to a power analysis:
 1. Pick a hypothetical effect size, e.g. $\mu_T - \mu_C = 0.05$
 2. Calculate the distribution of T under that effect size.
 3. Compute the probability of rejecting the null under that distribution.
 4. Repeat for different effect sizes.

Power Graph

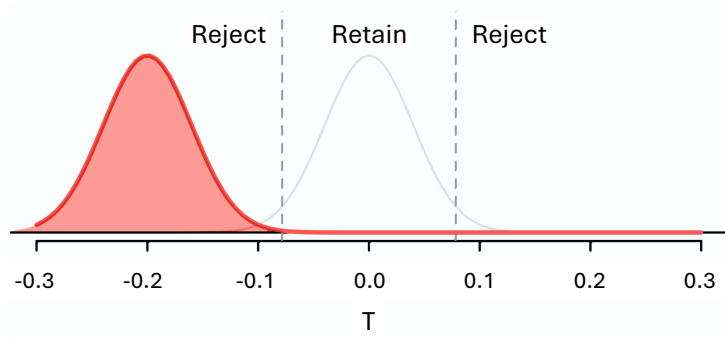


Power Graph



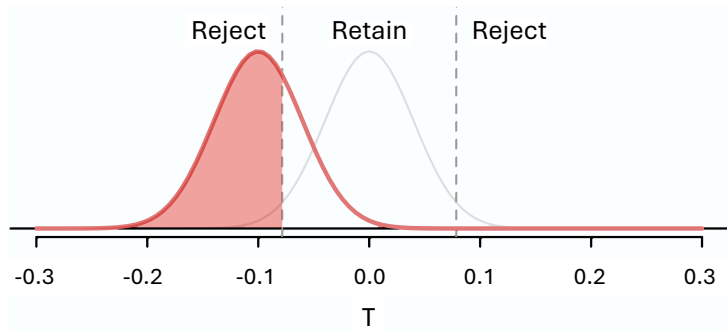
Assumed treatment effect = 0.05 and power = 0.24.

Power Graph



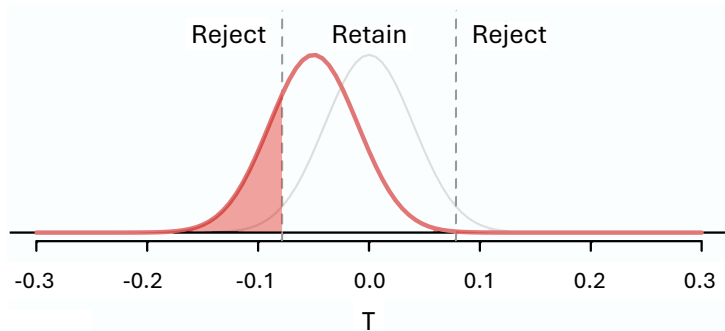
Assumed treatment effect = -0.2 and power = 0.999.

Power Graph



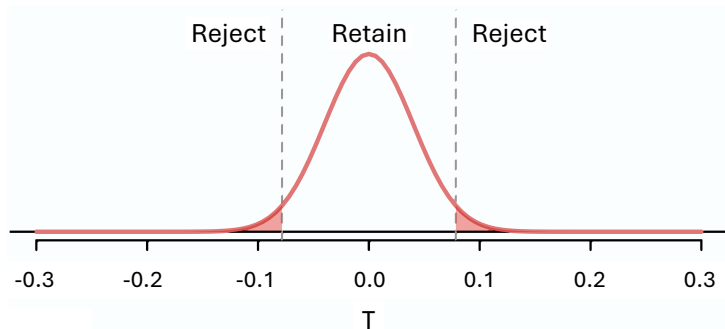
Assumed treatment effect = -0.1 and power = 0.705.

Power Graph



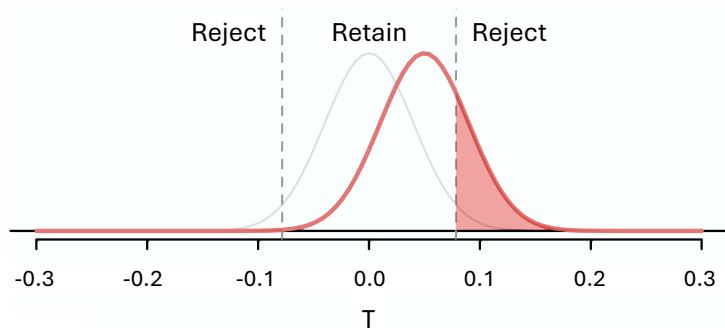
Assumed treatment effect = -0.05 and power = 0.24 .

Power Graph



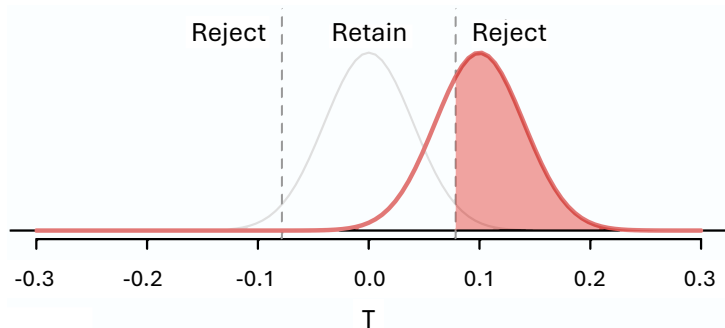
Assumed treatment effect = 0 and power = 0.05.

Power Graph



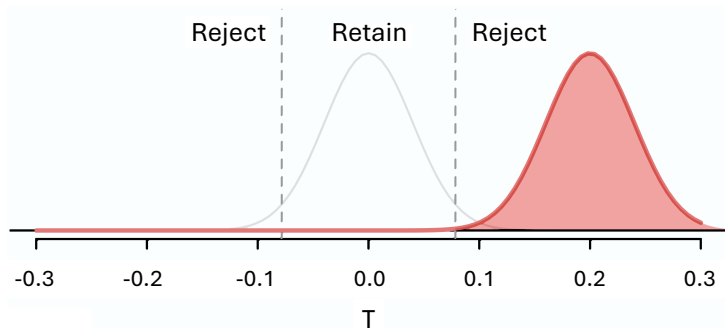
Assumed treatment effect = 0.05 and power = 0.24.

Power Graph



Assumed treatment effect = 0.1 and power = 0.705.

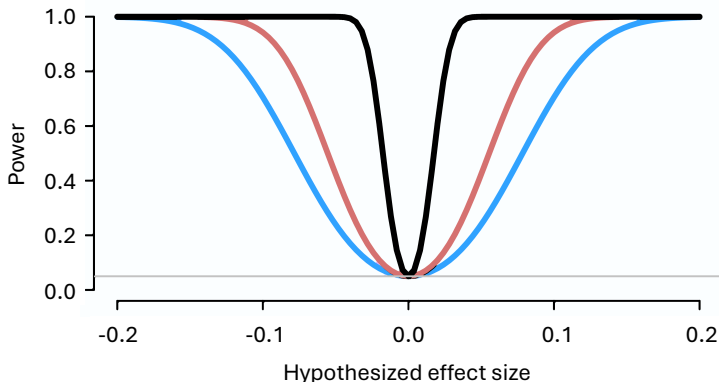
Power Graph



Assumed treatment effect = 0.2 and power = 0.999.

A Power Analysis

- We can calculate the power for every possible effect size and plot the resulting **power curve**:
 - $n = 500$ (blue), 1000 (red), 10,000 (black)

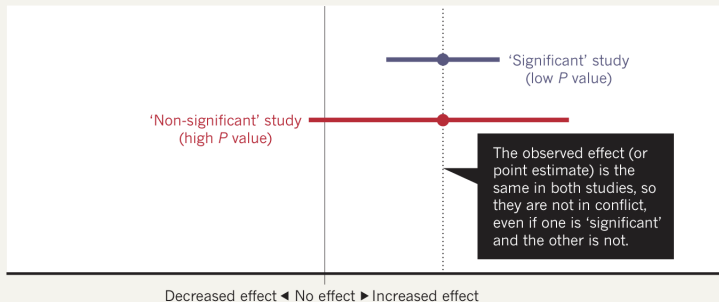


5/ Issues with Hypothesis Testing

Significant vs. Not Significant

BEWARE FALSE CONCLUSIONS

Studies currently dubbed 'statistically significant' and 'statistically non-significant' need not be contradictory, and such designations might cause genuine effects to be dismissed.



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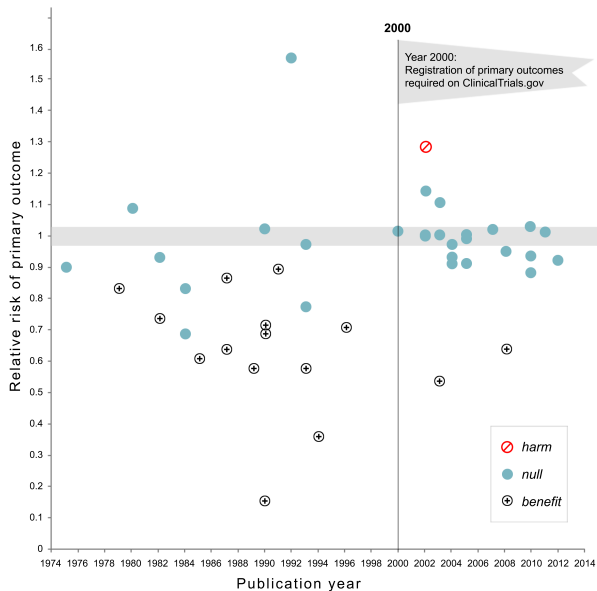
Source: "Scientists rise up against statistical significance" (Nature, 2019)

What Kind of Significance?

There are different types of significance that don't all have to be true together:

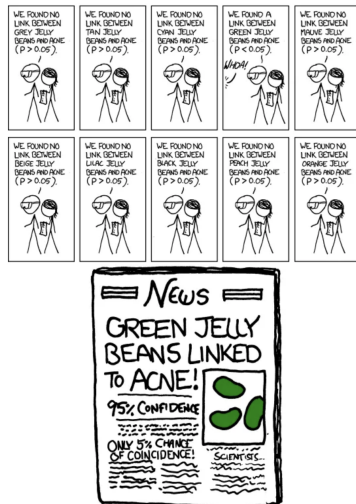
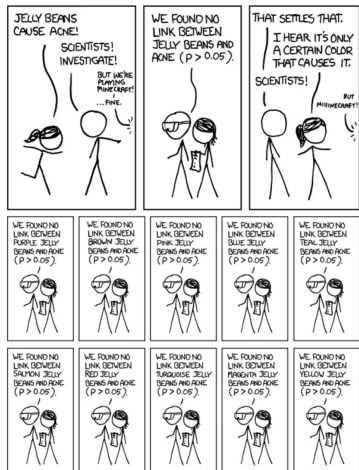
1. **Statistical significance:** we can reject the null of no effect.
2. **Causal significance:** we can interpret our estimated difference in means as a causal effect.
3. **Practical significance:** the estimated effect is meaningfully large (magnitude).

'Chart of the Decade' (Kaplan and Irvin, 2015)



P-Hacking

Performing a large number of statistical analyses in different ways with the intention of finding a desirable p-value (e.g., lower than 0.05).



Reporting Convention



Change in Journal of Marketing's Policy for Reporting Results

2.11.2025



From March 1, 2025, onward, the *Journal of Marketing* (JM) will adopt a new policy for reporting empirical results. This policy will hold for new submissions and revisions and is based on the recommendations for comprehensive reporting of statistical results as laid out in the article by [McShane et al. \(2024\)](#), and many articles from other fields such as strategy, biology, and medicine.

In addition to the guidelines currently available on the journal's [submission guidelines page](#) and on the [data transparency policy page](#), JM will implement the following new reporting guidelines:



1. JM submissions should report the actual p -values (three digits) rather than the threshold p -value in tables, running text, and in hypothesis testing.
2. Submissions should refrain from adding asterisks that signify thresholds (e.g., $p < .05$, $**p < .01$) to tables. Rather than using wording such as "the effect of X on Y is significant at .05," report the effect with the p -value in parentheses. For example, "the effect of X on Y is .20 ($p = .047$)."¹ Following a convention in experimental research, the p -values would be reported as (for example, " $T(1, 200) = 5.27, p = .023$ ").
3. Parameter estimates in tables should include the standard errors.
4. Effect sizes need to be reported. This can be achieved by adding corresponding information

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P-hacking and Significance Stars

Costanza Naguib

The Economic Journal, ueaf073, <https://doi.org/10.1093/ej/ueaf073>

Published: 27 August 2025 Article history

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Abstract

In mid-2016, all journals of the American Economic Association stopped including significance stars in their regression tables. This policy aimed to reduce the emphasis on statistical significance and shift the focus to the broader economic importance of research findings. This study examines the impact of this change on p -hacking and publication bias. I employ a difference-in-difference design, as well as the battery of tests proposed by Elliott et al. (2022b). I find no statistically significant effect of this policy.

JEL: A11 · Role of Economics; Role of Economists; Market for Economists, A14 · Sociology of Economics, C13 · Estimation: General

Issue Section: Article

Enjoy Your Weekends! :)

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